Alternative Data Research on Datadog

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I. Data selection and Rationale

Eight major financial and operating metrics reported by Datadog every quarter were chosen for this study due to their consistent availability in the past 12 quarters. They are revenue, billings, billings, remaining performance obligations (RPO), number of large customers—defined as customers who provide annual run-rate revenue (ARR) of at least $100,000— and the quarter-over-quarter (QoQ) and year-over-year (YoY) growth rates of each of them. This report proposes three alternative data sources that could act as leading indicators of the eight quarterly metrics. The justifications of each proposal are elucidated in the rest of the section.

PayPal stock returns (NASDAQ: PYPL)

Datadog's consumption-based pricing [1] suggests that the evolution of the cloud observability and security needs of its clients would lead to measurable changes in the quarterly performance metrics. The steady increase in proportion of the total ARR being large customers from 78% to 90% between FY2020 and FY2025 [2] further shows that large customers represent an increasingly important component of Datadog's revenue.

As of April 2026, the financial services industry accounts for the biggest share of the client base of Datadog at 26.04% [3]. Companies operating in the United States make up the majority of all its customers at 77.5% [3]. The demand from an American financial services firm is therefore more likely to affect the business performance of Datadog compared to firms in other industries and countries. The most direct measure of a client's ongoing demand for the Datadog platform would be the client's reported earnings. However, this information is released quarterly at approximately the same time as Datadog's own earnings, so it cannot be relied on as an advance estimate of the business performance of Datadog.

An indirect but more responsive indicator of a client's usage of the platform is the client's stock price, assuming the financial services offered by the client require the platform. Changes in stock prices reflect the market sentiment of the client's value, informed by the ongoing and predicted demand for the client's financial services. Rising or falling demand would affect the amount the client uses the platform and as a result, the earnings Datadog makes from the client.

One American financial services company that makes a promising candidate whose stock prices can serve as a predictor of Datadog's business performance is PayPal. PayPal has been verified to be a client and its operation being an online payments system, makes its usage of the Datadog platform likely to be substantial. This motivates the hypothesis that movements in PayPal's stock price could be a leading indicator of Datadog's business performance.

PayPal historical daily stock prices from 12 August 2016 to 11 August 2026 were obtained for free from NASDAQ and used to compute returns. The return r_t of an asset is the change in price over time interval Δt, as in

$$r_t = \frac{p_t - p_{t-\Delta t}}{p_{t-\Delta t}}. \tag{1}$$

with p_t being the price at time t. Data becomes available every market day at closing time. One limitation is that it may not be publicly known if PayPal eventually stops using the Datadog platform, and after that occurs the effect that PYPL returns would have on Datadog may become negligible. To mitigate that, the aggregated returns of major financial services clients could be utilized as a leading indicator instead.

Thematic sentiment score of technology news headlines

Among the numerous risk factors to financial and operation results identified by Datadog [4], many may be detected in non-confidential information: a disadvantageous climate in the economy or the cloud observability industry, diminished information technology budgets, decline in spending on Datadog products by customers, customers' decisions against renewing or expanding subscriptions, and Datadog or third-party service providers being unable to prevent cybersecurity or data privacy breaches. A common data source that could signal the appearance of these risk factors would be technology news. For example, reports of the advent of new vulnerabilities or opportunities of cloud usage could influence existing and prospective customers' decisions to accelerate or slow down their scaling of cloud adoption, resulting in changes in customer consumption of the Datadog platform.

News headlines from the CNBC technology section from 19 September 2019 to 30 June 2026 were scored by SetFit NLP model according to their sentiments, into three themes in the context of cloud usage: "Threats and vulnerabilities", "Opportunities for growth", and "Neutral". The 30 most recent headlines within the dataset were scraped from the CNBC technology news RSS feed and older ones were scraped from an archived snapshot of the CNBC technology news section with Wayback Machine CDX API. One snapshot per month was selected and scraped. Data is available continuously. One limitation is that Datadog's own business performance or announcements could affect the technology news headlines, so that a correlation between the sentiment score and Datadog's metrics does not necessarily mean that the news indicate changes in Datadog's performance. To mitigate this, we may want to exclude headlines about Datadog.

Frequency of Datadog's product releases

As the main service Datadog provides to customers is observability, allowing them to identify and resolve problems across applications, customers require a myriad of features to be able to conduct their operations on Datadog [5,6]. Datadog's ability to implement a large number of feature integrations on a single ecosystem has been its edge over competitors in attracting new business [6,7]. Therefore, the introduction of new features to the platform could improve Datadog's future earnings, suggesting that the frequency of product releases may be a leading indicator of the eight quarterly metrics of interest.

Press announcements from 20 November 2012 to 30 June 2026 were scraped from the Datadog Newsroom website. By keyword classification, press announcements about product releases were identified. Data is available continuously. One possible bias that could arise is that keyword-based identification may incorrectly classify or miss product releases that use terminology unexpected at the time of development. One way to mitigate this would be to use an NLP binary classifier model to recognize product release notices among the press releases.

II. Methodology

Lead-lag analysis

The Hayashi-Yoshida (HY) estimator [8,9,10] is a measure of correlation between two time series of differing lengths and non-uniform intervals. It is ratio of the sum of products of differences in the two time-series X_t and Y_t whenever their time-intervals overlap, to the integrated covariance between the two time series,

$$\hat{\rho}(\ell) = \frac{\sum_{i,j} \Delta X(I_i^X)\Delta Y(I_j^Y)\mathbb{1}_{\{I_i^X \cap I_j^Y \neq 0\}}}{\sqrt{\sum_i [\Delta X(I_i^X)]^2 \sum_j \left[\Delta Y\left(I_j^Y\right)\right]^2}} \tag{2}$$

where the time-intervals between two datapoints in X_t and Y_t and are represented by $I_i^X = (t_{i-1}^X, t_i^X)$ and $I_j^Y = \left(t_{j-1}^Y, t_j^Y\right)$ respectively. During overlapping time-intervals, $\mathbb{1}_{\{I_i^X \cap I_{j+\ell}^Y \neq 0\}}$ equals 1, and 0 at other times. The lead-lag relationship between two time-series X_t and Y_t by studied artificially shifting the times of Y_t by a lag ℓ [9,10], and calculating the modified HY estimator,

$$\hat{\rho}(\ell) = \frac{\sum_{i,j} \Delta X(I_i^X)\Delta Y(I_j^Y)\mathbb{1}_{\{I_i^X \cap I_{j+\ell}^Y \neq 0\}}}{\sqrt{\sum_i [\Delta X(I_i^X)]^2 \sum_j \left[\Delta Y\left(I_j^Y\right)\right]^2}} \tag{3}$$

with $Y_{t+\ell}$ is the shifted time-series. If $l > 0$, $Y_{t+\ell}$ is shifted to the future and $\hat{\rho}(\ell)$ gives the extent of leadership that Y_t has over X_t at l. If $l < 0$, $Y_{t+\ell}$ is shifted to the past and $\hat{\rho}(\ell)$ gives the extent of leadership that X_t has over Y_t at l. Evaluating the HY estimator over a grid of lags, the l at which the absolute value of the HY estimator is maximum, known as the lead-lag time ℓ^*, signifies that Y_t leads if $\ell^* > 0$ or lags if $\ell^* < 0$ [9]. To determine if our alternative datasets could act as leading indicators to the metrics reported quarterly by Datadog, we compute the HY estimator over a two-dimensional grid of n days by l days to find the ideal length n of rolling average of alternative data values as well as the ideal lag l on the quarterly metrics such that ℓ^* is maximized. A range of 0 to 90 days was chosen for n while ℓ ranged from -90 to +90 days.

Predictive model

The pairs of alternative dataset and quarterly metric where the lead-lag analysis suggested the alternative dataset was lagging behind the quarterly metric were excluded from the predictive model. A chronological train-test split was made on the alternative dataset time-series in a 70-30 ratio. Each pair of alternative dataset and quarterly metric was trained on a linear regression model and a random forest regression model at the ideal rolling window length n and ideal lag l on the quarterly metric, to find an appropriate model that is able to make accurate predictions. The R^2 and RMSE scores of both models were assessed. The random forest regression models did not exhibit better accuracy than linear regression and runs a risk of overfitting considering the small sample size of less than 20 observations, so the rest of the analysis was done on the linear regression predictions.

Out-of-sample validation

Within each type of quarterly metric (ie. between the raw metric, its QoQ growth rate, and YoY growth rates), we chose the pair of alternative datasets and the metric giving the highest R^2 score to be subjected to a walk-forward backtest in order to validate the accuracy of our linear-regression predictive model out-of-sample (OOS). As raw RPO and Billings were observed to move cyclically every year, the YoY growth rates were chosen regardless of their R^2 scores to eliminate the influence of cyclical factors in our analysis. The MAPE, RMSE, directional hit rate, and average risk-to-reward ratio were evaluated.

III. Findings

- The maximum HY estimator was generally found in long windows of rolling averages of PayPal returns and product releases frequency, ie. longer than 45 days, suggests a high signal-to-noise ratio in Paypal returns and a longer time needed for product releases to accumulate and have a statistically significant influence on the reported quarterly metrics.
- None of the pairs of alternative dataset and quarterly metric used to build a linear regression predictive model yielded an R^2 score above 0. This shows that there is no linear correlation between the alternative datasets selected in this study and the quarterly KPIs reported by Datadog at the window length n and lag l corresponding to $\hat{\rho}(\ell)$. There may be non-linear correlations present instead, considering that the absolute $\hat{\rho}(\ell)$ had moderate values (0.1 to 0.4).
- A 90-day rolling average of PayPal returns with a 77-day lead over the RPO YoY growth rate exhibited the highest directional hit rate at 69.23%. However, its average risk-to-reward ratio is also the largest. A trader who is willing to take a large financial risk is likely to profit from a trading strategy built on this pair of leading and lagging measures.

IV. Limitations

- HY estimator had not been tested on ultra-low frequency time-series ie. time-intervals longer than 60 seconds. More rigorous experimentation is needed to verify the accuracy of the HY estimator at measuring lead-lag relationships than this study.
- The analysis was restricted to the certain window lengths and lag periods, so longer accumulation periods or delayed effects may have been missed.
- This study focused on linear relationships between the alternative dataset and lagged quarterly metric, but alternative datasets may still have predictive power through non-linear relationships.

V. Bibliography

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VI. Appendix

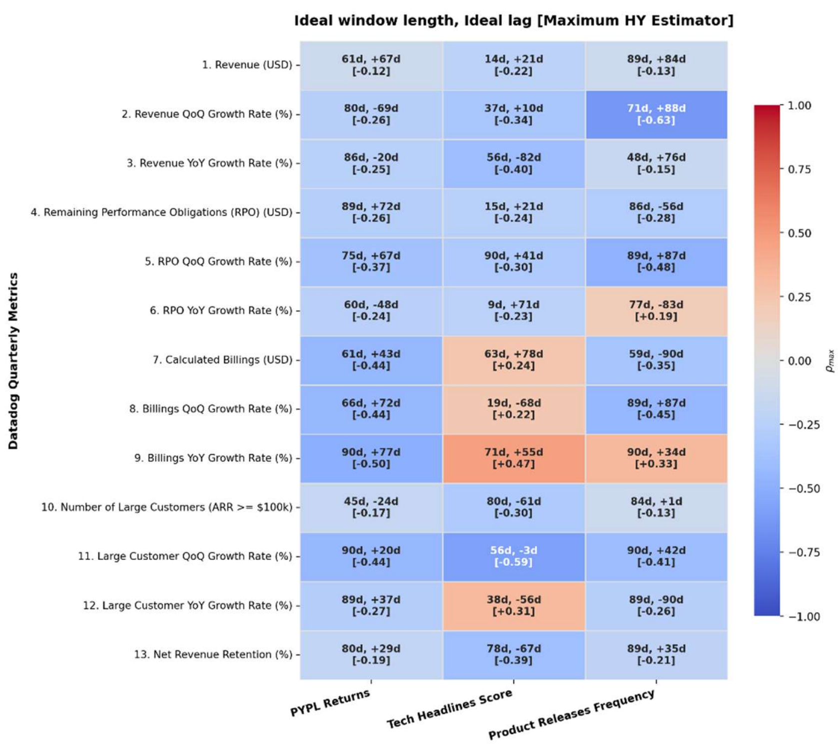

Figure A.1: Maximum HY estimator $\hat{\rho}(\ell)$ of each pair of alternative data and quarterly metric. $\hat{\rho}(\ell)$ takes values between -1 (blue) and 1 (red), indicating the magnitude of correlations between two time-series. A positive lag means the alternative dataset leads the quarterly metric whereas a negative lag means the quarterly metric leads the alternative dataset.

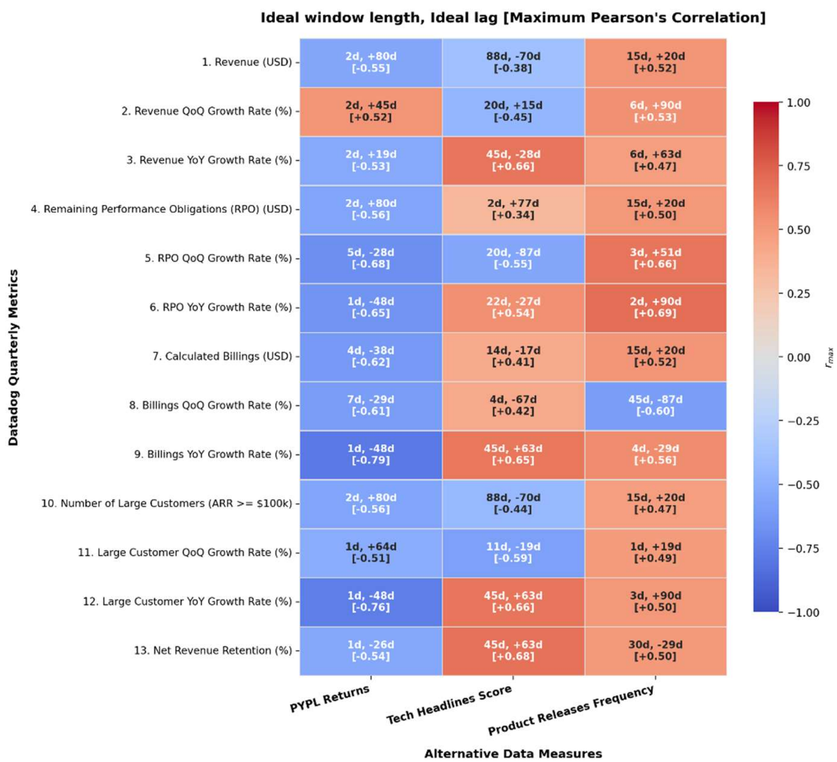

Figure A.2: Maximum Pearson's correlation coefficient $r(\ell)$ of each pair of alternative data and quarterly metric.

Dataset	Metric	MAPE (%)	RMSE	Directional hit rate (%)	Average R/R ratio	OOS quarters tested
Product Releases Frequency	Revenue QoQ Growth Rate (%)	96.91	5.75	57.89	2.37	19
Tech Headlines Score	RPO YoY Growth Rate (%)	64.38	26.57	57.14	1.89	14
PYPL Returns	Billings YoY Growth Rate (%)	92.39	24.33	69.23	4.98	13
PYPL Returns	Large Customer QoQ Growth Rate (%)	153.2	5.19	41.18	3.29	17

Table A.1: OOS walk-forward backtesting results.